

```
# =====
# EMPLOYEE ATTRITION - PROJECT PLANNING
# Google Colab
# =====

!pip -q install pandas numpy matplotlib seaborn openpyxl

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
from google.colab import files
from IPython.display import display, Image

print("Libraries imported successfully!")
```

Libraries imported successfully!

```
# Upload HR Attrition CSV file

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

print("Uploaded file:", file_name)

df = pd.read_csv(file_name)

print("Dataset loaded successfully!")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

display(df.head())
```

archive (9).zip

archive (9).zip(application/x-zip-compressed) - 51314 bytes, last modified: 8/30/2026 - 100% done

Saving archive (9).zip to archive (9).zip

Uploaded file: archive (9).zip

Dataset loaded successfully!

Rows: 1470

Columns: 35

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeNumber
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	

5 rows × 35 columns

```
print("=" * 60)
print("DATASET INFORMATION")
print("=" * 60)

print("\nDataset Shape:")
```

```
print(df.shape)

print("\nColumn Names:")
for i, col in enumerate(df.columns, 1):
    print(i, "-", col)

print("\nData Types:")
display(df.dtypes)

print("\nFirst 5 Rows:")
display(df.head())

print("\nLast 5 Rows:")
display(df.tail())
```


=====

DATASET INFORMATION

=====

Dataset Shape:
(1470, 35)

- Column Names:
- 1 - Age
 - 2 - Attrition
 - 3 - BusinessTravel
 - 4 - DailyRate
 - 5 - Department
 - 6 - DistanceFromHome
 - 7 - Education
 - 8 - EducationField
 - 9 - EmployeeCount
 - 10 - EmployeeNumber
 - 11 - EnvironmentSatisfaction
 - 12 - Gender
 - 13 - HourlyRate
 - 14 - JobInvolvement
 - 15 - JobLevel
 - 16 - JobRole
 - 17 - JobSatisfaction
 - 18 - MaritalStatus
 - 19 - MonthlyIncome
 - 20 - MonthlyRate
 - 21 - NumCompaniesWorked
 - 22 - Over18
 - 23 - OverTime
 - 24 - PercentSalaryHike
 - 25 - PerformanceRating
 - 26 - RelationshipSatisfaction
 - 27 - StandardHours
 - 28 - StockOptionLevel
 - 29 - TotalWorkingYears
 - 30 - TrainingTimesLastYear
 - 31 - WorkLifeBalance
 - 32 - YearsAtCompany
 - 33 - YearsInCurrentRole
 - 34 - YearsSinceLastPromotion
 - 35 - YearsWithCurrManager

Data Types:

	0
Age	int64
Attrition	object
BusinessTravel	object
DailyRate	int64
Department	object
DistanceFromHome	int64
Education	int64
EducationField	object
EmployeeCount	int64
EmployeeNumber	int64
EnvironmentSatisfaction	int64
Gender	object
HourlyRate	int64
JobInvolvement	int64

JobLevel	int64
JobRole	object
JobSatisfaction	int64
MaritalStatus	object
MonthlyIncome	int64
MonthlyRate	int64
NumCompaniesWorked	int64
Over18	object
OverTime	object
PercentSalaryHike	int64
PerformanceRating	int64
RelationshipSatisfaction	int64
StandardHours	int64
StockOptionLevel	int64
TotalWorkingYears	int64
TrainingTimesLastYear	int64
WorkLifeBalance	int64
YearsAtCompany	int64
YearsInCurrentRole	int64
YearsSinceLastPromotion	int64
YearsWithCurrManager	int64

dtype: object

First 5 Rows:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	Emp
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	

5 rows × 35 columns

Last 5 Rows:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	
1465	36	No	Travel_Frequently	884	Research & Development	23	2	Medical	
1466	39	No	Travel_Rarely	613	Research & Development	6	1	Medical	
1467	27	No	Travel_Rarely	155	Research & Development	4	3	Life Sciences	
1468	49	No	Travel_Frequently	1023	Sales	2	3	Medical	

```
# Missing value analysis

missing_values = pd.DataFrame({
    "Column": df.columns,
    "Missing_Count": df.isnull().sum(),
    "Missing_Percentage": (df.isnull().sum() / len(df) * 100).round(2)
})

missing_values = missing_values.sort_values(
    by="Missing_Count",
    ascending=False
)

display(missing_values)

print("Total missing values:", df.isnull().sum().sum())
```

	Column	Missing_Count	Missing_Percentage	
Age	Age	0	0.0	
Attrition	Attrition	0	0.0	
BusinessTravel	BusinessTravel	0	0.0	
DailyRate	DailyRate	0	0.0	
Department	Department	0	0.0	
DistanceFromHome	DistanceFromHome	0	0.0	
Education	Education	0	0.0	
EducationField	EducationField	0	0.0	
EmployeeCount	EmployeeCount	0	0.0	
EmployeeNumber	EmployeeNumber	0	0.0	
EnvironmentSatisfaction	EnvironmentSatisfaction	0	0.0	
Gender	Gender	0	0.0	
HourlyRate	HourlyRate	0	0.0	
JobInvolvement	JobInvolvement	0	0.0	
JobLevel	JobLevel	0	0.0	
JobRole	JobRole	0	0.0	
JobSatisfaction	JobSatisfaction	0	0.0	
MaritalStatus	MaritalStatus	0	0.0	
MonthlyIncome	MonthlyIncome	0	0.0	
MonthlyRate	MonthlyRate	0	0.0	
NumCompaniesWorked	NumCompaniesWorked	0	0.0	
Over18	Over18	0	0.0	
OverTime	OverTime	0	0.0	
PercentSalaryHike	PercentSalaryHike	0	0.0	
PerformanceRating	PerformanceRating	0	0.0	
RelationshipSatisfaction	RelationshipSatisfaction	0	0.0	
StandardHours	StandardHours	0	0.0	
StockOptionLevel	StockOptionLevel	0	0.0	
TotalWorkingYears	TotalWorkingYears	0	0.0	
TrainingTimesLastYear	TrainingTimesLastYear	0	0.0	
WorkLifeBalance	WorkLifeBalance	0	0.0	
YearsAtCompany	YearsAtCompany	0	0.0	
YearsInCurrentRole	YearsInCurrentRole	0	0.0	
YearsSinceLastPromotion	YearsSinceLastPromotion	0	0.0	
YearsWithCurrManager	YearsWithCurrManager	0	0.0	

Total missing values: 0

Next steps:

[Generate code with missing_values](#)

[New interactive sheet](#)

```
# Identify target variable
```

```
target = "Attrition"

if target in df.columns:
    print("Target variable found:", target)
else:
    print("Target variable not found.")

print("\nTarget Distribution:")
display(df[target].value_counts())

print("\nTarget Percentage:")
display(
    df[target].value_counts(normalize=True).mul(100).round(2)
)
```

Target variable found: Attrition

Target Distribution:

	count
Attrition	
No	1233
Yes	237

dtype: int64

Target Percentage:

	proportion
Attrition	
No	83.88
Yes	16.12

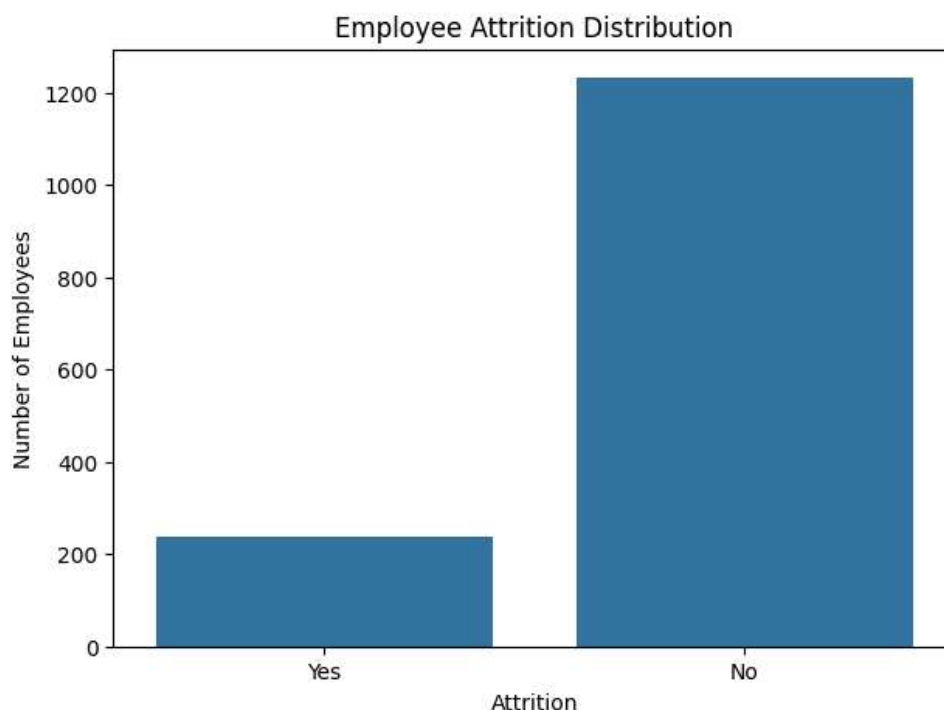
dtype: float64

```
plt.figure(figsize=(7,5))

sns.countplot(
    data=df,
    x="Attrition"
)

plt.title("Employee Attrition Distribution")
plt.xlabel("Attrition")
plt.ylabel("Number of Employees")

plt.show()
```

```
# =====
# BUSINESS KPI CALCULATION
# =====

total_employees = len(df)

employees_left = (df["Attrition"] == "Yes").sum()

employees_stayed = (df["Attrition"] == "No").sum()

attrition_rate = employees_left / total_employees * 100

retention_rate = employees_stayed / total_employees * 100

print("=" * 50)
print("EMPLOYEE ATTRITION KPIs")
print("=" * 50)

print(f"Total Employees      : {total_employees}")
print(f"Employees Left         : {employees_left}")
print(f"Employees Stayed        : {employees_stayed}")
print(f"Attrition Rate           : {attrition_rate:.2f}%")
print(f"Retention Rate           : {retention_rate:.2f}%")
```

```
=====
EMPLOYEE ATTRITION KPIs
=====
Total Employees      : 1470
Employees Left       : 237
Employees Stayed     : 1233
Attrition Rate        : 16.12%
Retention Rate        : 83.88%
```

```
# Numerical features
numerical_features = df.select_dtypes(
    include=["int64", "float64"]
).columns.tolist()

# Categorical features
categorical_features = df.select_dtypes(
```

```

        include=["object", "category"]
    ).columns.tolist()

    print("Numerical Features:")
    print(numerical_features)

    print("\nCategorical Features:")
    print(categorical_features)

    print("\nNumber of Numerical Features:",
          len(numerical_features))

    print("Number of Categorical Features:",
          len(categorical_features))

```

Numerical Features:
 ['Age', 'DailyRate', 'DistanceFromHome', 'Education', 'EmployeeCount', 'EmployeeNumber', 'EnvironmentSatisfac

Categorical Features:
 ['Attrition', 'BusinessTravel', 'Department', 'EducationField', 'Gender', 'JobRole', 'MaritalStatus', 'Over1

Number of Numerical Features: 26
 Number of Categorical Features: 9

```

# =====
# FEATURE DICTIONARY
# =====

feature_dictionary = pd.DataFrame({
    "Feature": df.columns,
    "Data Type": df.dtypes.astype(str).values,
    "Unique Values": [df[col].nunique() for col in df.columns],
    "Missing Values": [df[col].isnull().sum() for col in df.columns],
})

# Add feature category
feature_dictionary["Feature Type"] = feature_dictionary["Feature"].apply(
    lambda x: "Numerical" if x in numerical_features else "Categorical"
)

display(feature_dictionary)

```

	Feature	Data Type	Unique Values	Missing Values	Feature Type	
0	Age	int64	43	0	Numerical	
1	Attrition	object	2	0	Categorical	
2	BusinessTravel	object	3	0	Categorical	
3	DailyRate	int64	886	0	Numerical	
4	Department	object	3	0	Categorical	
5	DistanceFromHome	int64	29	0	Numerical	
6	Education	int64	5	0	Numerical	
7	EducationField	object	6	0	Categorical	
8	EmployeeCount	int64	1	0	Numerical	
9	EmployeeNumber	int64	1470	0	Numerical	
10	EnvironmentSatisfaction	int64	4	0	Numerical	
11	Gender	object	2	0	Categorical	
12	HourlyRate	int64	71	0	Numerical	
13	JobInvolvement	int64	4	0	Numerical	
14	JobLevel	int64	5	0	Numerical	
15	JobRole	object	9	0	Categorical	
16	JobSatisfaction	int64	4	0	Numerical	
17	MaritalStatus	object	3	0	Categorical	
18	MonthlyIncome	int64	1349	0	Numerical	
19	MonthlyRate	int64	1427	0	Numerical	
20	NumCompaniesWorked	int64	10	0	Numerical	
21	Over18	object	1	0	Categorical	
22	OverTime	object	2	0	Categorical	
23	PercentSalaryHike	int64	15	0	Numerical	
24	PerformanceRating	int64	2	0	Numerical	
25	RelationshipSatisfaction	int64	4	0	Numerical	
26	StandardHours	int64	1	0	Numerical	
27	StockOptionLevel	int64	4	0	Numerical	
28	TotalWorkingYears	int64	40	0	Numerical	
29	TrainingTimesLastYear	int64	7	0	Numerical	
30	WorkLifeBalance	int64	4	0	Numerical	
31	YearsAtCompany	int64	37	0	Numerical	
32	YearsInCurrentRole	int64	19	0	Numerical	
33	YearsSinceLastPromotion	int64	16	0	Numerical	
34	YearsWithCurrManager	int64	18	0	Numerical	

Next steps:

[Generate code with feature_dictionary](#)

[New interactive sheet](#)

```
dataset_report = pd.DataFrame({
    "Metric": [
        "Number of Rows",
```

```
        "Number of Columns",
        "Numerical Features",
        "Categorical Features",
        "Missing Values",
        "Duplicate Rows",
        "Target Variable",
        "Employees Left",
        "Employees Stayed",
        "Attrition Rate",
        "Retention Rate"
    ],

    "Value": [
        df.shape[0],
        df.shape[1],
        len(numerical_features),
        len(categorical_features),
        df.isnull().sum().sum(),
        df.duplicated().sum(),
        target,
        employees_left,
        employees_stayed,
        f"{attrition_rate:.2f}%",
        f"{retention_rate:.2f}%"
    ]
})

display(dataset_report)

dataset_report.to_csv(
    "dataset_report.csv",
    index=False
)
```

	Metric	Value	
0	Number of Rows	1470	
1	Number of Columns	35	
2	Numerical Features	26	
3	Categorical Features	9	
4	Missing Values	0	
5	Duplicate Rows	0	
6	Target Variable	Attrition	
7	Employees Left	237	
8	Employees Stayed	1233	
9	Attrition Rate	16.12%	
10	Retention Rate	83.88%	

Next steps:

[Generate code with dataset_report](#)

[New interactive sheet](#)

```
print("Statistical Summary of Numerical Features")

display(
    df[numerical_features].describe().T
)
```

Statistical Summary of Numerical Features									
	count	mean	std	min	25%	50%	75%	max	
Age	1470.0	36.923810	9.135373	18.0	30.00	36.0	43.00	60.0	
DailyRate	1470.0	802.485714	403.509100	102.0	465.00	802.0	1157.00	1499.0	
DistanceFromHome	1470.0	9.192517	8.106864	1.0	2.00	7.0	14.00	29.0	
Education	1470.0	2.912925	1.024165	1.0	2.00	3.0	4.00	5.0	
EmployeeCount	1470.0	1.000000	0.000000	1.0	1.00	1.0	1.00	1.0	
EmployeeNumber	1470.0	1024.865306	602.024335	1.0	491.25	1020.5	1555.75	2068.0	
EnvironmentSatisfaction	1470.0	2.721769	1.093082	1.0	2.00	3.0	4.00	4.0	
HourlyRate	1470.0	65.891156	20.329428	30.0	48.00	66.0	83.75	100.0	
JobInvolvement	1470.0	2.729932	0.711561	1.0	2.00	3.0	3.00	4.0	
JobLevel	1470.0	2.063946	1.106940	1.0	1.00	2.0	3.00	5.0	
JobSatisfaction	1470.0	2.728571	1.102846	1.0	2.00	3.0	4.00	4.0	
MonthlyIncome	1470.0	6502.931293	4707.956783	1009.0	2911.00	4919.0	8379.00	19999.0	
MonthlyRate	1470.0	14313.103401	7117.786044	2094.0	8047.00	14235.5	20461.50	26999.0	
NumCompaniesWorked	1470.0	2.693197	2.498009	0.0	1.00	2.0	4.00	9.0	
PercentSalaryHike	1470.0	15.209524	3.659938	11.0	12.00	14.0	18.00	25.0	
PerformanceRating	1470.0	3.153741	0.360824	3.0	3.00	3.0	3.00	4.0	
RelationshipSatisfaction	1470.0	2.712245	1.081209	1.0	2.00	3.0	4.00	4.0	
StandardHours	1470.0	80.000000	0.000000	80.0	80.00	80.0	80.00	80.0	
StockOptionLevel	1470.0	0.793878	0.852077	0.0	0.00	1.0	1.00	3.0	
TotalWorkingYears	1470.0	11.279592	7.780782	0.0	6.00	10.0	15.00	40.0	
TrainingTimesLastYear	1470.0	2.799320	1.289271	0.0	2.00	3.0	3.00	6.0	
WorkLifeBalance	1470.0	2.761224	0.706476	1.0	2.00	3.0	3.00	4.0	
YearsAtCompany	1470.0	7.008163	6.126525	0.0	3.00	5.0	9.00	40.0	
YearsInCurrentRole	1470.0	4.229252	3.623137	0.0	2.00	3.0	7.00	18.0	
YearsSinceLastPromotion	1470.0	2.187755	3.222430	0.0	0.00	1.0	3.00	15.0	
YearsWithCurrManager	1470.0	4.123129	3.568136	0.0	2.00	3.0	7.00	17.0	

```
# Plot distributions for numerical features

for feature in numerical_features[:10]:

    plt.figure(figsize=(7,4))

    sns.histplot(
        data=df,
        x=feature,
        kde=True
    )

    plt.title(f"Distribution of {feature}")
    plt.xlabel(feature)
    plt.ylabel("Frequency")

    plt.show()
```

